Unit II :: Mathematical Typesetting and Technical Elements in LaTeX

SEC-03-PHY-91-R-03

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Syllabus

Unit II: Mathematical Typesetting and Technical Elements in LaTeX

Math mode: inline and display modes; math fonts: mathcal, mathbb, mathds, mathfrak; superscripts and subscripts, pre-defined functions, fractions, roots, Greek letters; integrals, summations and products; mathematical grids: multi-line formulae, gathering system of equations, aligning system of equations, matrices; LaTeX packages for Physics: siunitx, physics; floats in LaTeX: table and figure environments; typesetting tables using tabular; formatting figures using includegraphics.

1 Overview of math mode in LaTeX

In LaTeX, the typesetting of mathematical expressions requires entering a dedicated operating environment known as *math mode*. In math mode, LaTeX does the following

- Treats alphabet symbols as mathematical variables.
- Typesets letters in italics by default.
- Ignores standard whitespace characters.
- Adjusts spacing between symbols dynamically based on their semantic classification, such as operators, relations and delimiters.

1.1 Inline math mode

Inline math mode is used to embed short mathematical expressions and variables directly within a sentence without disrupting the paragraph flow. We can typeset a mathematical expression in inline mode using any of the following syntax:

1. Enclosing the expression within `$...$`, for example `$F=ma$`.
2. Enclosing the expression within `\(...\)`, for example `\(p=mv\)`.
3. Placing the expression within the `math` environment, for example
```
\begin{math}
    U=mgh
\end{math}
```

For instance, the following code contains all of the above three syntax

```
In classical mechanics, an object's translational motion and energetic state
near Earth's surface can be characterized by Newton's second law of motion $F=
ma$, linear momentum \(p=mv\), and gravitational potential energy
\begin{math}
  U=mgh
\end{math}.
```

which produces the following output:

In classical mechanics, an object's translational motion and energetic state near Earth's surface can be characterized by Newton's second law of motion $F = ma$, linear momentum $p = mv$, and gravitational potential energy $U = mgh$.

1.2 Display math mode

Display math mode is used for prominent, complex or multi-line equations that require dedicated vertical separation, centered alignment and full symbol scaling.

For display mode, we have two types of syntax:

1. *Un-numbered display syntax*:

   (a) Enclosing the expression within `$$...$$`, for example `$$F=ma$$`.

   (b) Enclosing the expression within `\[...\]`, for example `\[p=mv\]`.

   (c) Placing the expression within the `displaymath` environment, for example
   ```
   \begin{displaymath}
       U=mgh
   \end{displaymath}
   ```

2. *Numbered display syntax*: Placing the expression within the `equation` environment, in which case the equation will also be given a number. For example
   ```
   \begin{equation}
       T=\frac{1}{2}mv^2
   \end{equation}
   ```
   Note: The numbering of the equation can be suppressed by using the `equation*` environment, with an asterisk (*) appended. For example
   ```
   \begin{equation*}
       E=mgh+\frac{1}{2}mv^2
   \end{equation*}
   ```

For instance, the following code contains all of the above three syntax

```
In classical mechanics, an object's translational motion and energetic state
near Earth's surface can be characterized by Newton's second law of motion $$F=
ma$$ linear momentum \[p=mv\] gravitational potential energy
\begin{displaymath}
  U=mgh
\end{displaymath}
and kinetic energy
\begin{equation}
  T=\frac{1}{2}mv^2
\end{equation}
with the mechanical energy
\begin{equation*}
  E=mgh+\frac{1}{2}mv^2
\end{equation*}
```

which produces the following output:

In classical mechanics, an object's translational motion and energetic state near Earth's surface can be characterized by Newton's second law of motion

$$F = ma$$

linear momentum

$$p = mv$$

gravitational potential energy

$$U = mgh$$

and kinetic energy

$$T = \frac{1}{2}mv^2 \tag{1}$$

with the mechanical energy

$$E = mgh + \frac{1}{2}mv^2$$

2 Mathematical typography in LaTeX

2.1 Superscripts and subscripts

Superscripts and subscripts are commonly used to represent exponents (such as 3^2, x^{-3}) and indexed quantities (such as m_1, K_0) respectively. In LaTeX, superscripts are typeset using the *caret* symbol (`^`) while subscripts are typeset using the *underscore* symbol (`_`).

If a superscript or subscript contains more than a single character, the entire exponent or index must be enclosed in braces `{ }`.

For instance, the following code contains expressions with exponents and indices

```
$$ax^2+bx+c=0$$

$$f(x)=e^{-ax^2}$$

$$m_1=5, m_{12}=36.5$$
```

which produces the output

$$a x^2 + b x + c = 0$$

$$f(x) = e^{-a x^2}$$

$$m_1 = 5, m_{12} = 36.5$$

2.2 Fractions and roots

In LaTeX, fractions are typeset using the `\frac` macro. The general syntax is

```
\frac{numerator}{denominator}
```

For fractions contained inline, where vertical space is limited, standard slash notation `a/b` or the `\tfrac{num}{den}` macro can be used.

For instance, the following code contains examples of typesetting fractions

```
The kinetic energy of a particle of mass $m$ and moving with speed $v$ is $\
frac{1}{2}mv^2$.

Using the law of equipartition of energy, the average energy of a monoatomic
molecule is obtained to be $\tfrac{3}{2}k_BT$.
```

which produces the output

The kinetic energy of a particle of mass m and moving with speed v is $\frac{1}{2}mv^2$.
Using the law of equipartition of energy, the average energy of a monoatomic molecule is obtained to be $\frac{3}{2}k_BT$.

LaTeX allows us to typeset the square root using the `\sqrt` macro which has the general syntax as

```
\sqrt{expression}
```

The `\sqrt` macro can be extended to typeset the nth root using an optional argument. The general syntax for this is

```
\sqrt[n]{expression}
```

For instance, the following code contains examples of typesetting roots

```
The root mean squared speed of molecules is $\sqrt{\dfrac{3k_BT}{m}}$, as per
Maxwell-Boltzmann distribution.

The roots of a quadratic equation are
\begin{equation}
  x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}
\end{equation}
```

which produces the output

The root mean squared speed of molecules is $\sqrt{\dfrac{3k_BT}{m}}$, as per Maxwell-Boltzmann distribution.
The roots of a quadratic equation are

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a} \tag{2}$$

2.3 Pre-defined functions and operators

Standard mathematical functions (such as trigonometric, logarithmic and limit operators) must not be typed as plain text in math mode; otherwise, LaTeX would render them in variable italics. For example, `sin(x)` typesets as a product of s, i and n. Hence to render this correctly in upright roman typeface with appropriate surrounding whitespace, LaTeX provides pre-defined operator macros for a bunch of functions as given below.

- *Trigonometric and hyperbolic functions*: `\sin`, `\cos`, `\tan`, `\arcsin`, `\arccos`, `\arctan`, `\sinh`, `\cosh`, `\tanh`
- *Logarithmic and exponential functions*: `\exp`, `\log`, `\ln`
- *Limits and optimization*: `\lim`, `\min`, `\max`, `\sup`, `\inf`
- *Linear algebra*: `\det`, `\ker`, `\deg`

2.4 Summation, product and integral operators

In LaTeX, the summation, product and integral operators are typeset using the \sum, \prod and \int macros. These are variable-sized operators and get rendered in different sizes, depending on whether they are in the inline or display modes.

The following code typesets these in the inline mode:

```
$\sum_{i=1}^{N}(2i-1)$, $\prod_{k=1}^{N}\frac{1}{2^k}$ and $\int_{0}^{5}e^{-x}
dx$
```

which gives the following output:

$\sum_{i=1}^{N}(2i-1)$, $\prod_{k=1}^{N}\frac{1}{2^k}$ and $\int_{0}^{5}e^{-x}dx$

In display math mode, variable-sized operators render larger than in inline mode, and their limits adjust automatically. The following code typesets the above expressions in the display mode:

```
$$\sum_{i=1}^{N}(2i-1)$$ $$\prod_{k=1}^{N}\frac{1}{2^k}$$ $$\int_{0}^{5}e^{-x}
dx$$
```

which gives the following output:

$$\sum_{i=1}^{N}(2i-1)$$

$$\prod_{k=1}^{N}\frac{1}{2^k}$$

$$\int_{0}^{5}e^{-x}dx$$

2.5 Greek letters

In physics, letters from the Greek alphabet are widely used to represent various physical quantities. In LaTeX, lowercase Greek letters are typeset by prefixing the letter's English name with a backslash, as given in the following table:

Letter	Macro	Letter	Macro	Letter	Macro	Letter	Macro
α	\alpha	η	\eta	ν	\nu	υ	\upsilon
β	\beta	θ	\theta	ξ	\xi	ϕ	\phi
γ	\gamma	ι	\iota	π	\pi	χ	\chi
δ	\delta	κ	\kappa	ρ	\rho	ψ	\psi
ϵ	\epsilon	λ	\lambda	σ	\sigma	ω	\omega
ζ	\zeta	μ	\mu	τ	\tau		

For the uppercase Greek letters that do not resemble any of the letters from the English alphabet, the typesetting is done in LaTeX by capitalizing the first letter of the macro for the same lowercase

letter. For instance, when the macro for the lowercase lambda λ, which is `\lambda`, is capitalized as `\Lambda`, we obtain the uppercase lambda Λ.

The following table provides the macros for the uppercase Greek letters.

Letter	Macro	Letter	Macro	Letter	Macro	Letter	Macro
Γ	`\Gamma`	Λ	`\Lambda`	Σ	`\Sigma`	Ψ	`\Psi`
Δ	`\Delta`	Ξ	`\Xi`	Υ	`\Upsilon`	Ω	`\Omega`
Θ	`\Theta`	Π	`\Pi`	Φ	`\Phi`		

2.6 Mathematical fonts

To represent specific mathematical structures (such as sets, vector spaces, or Lie algebras), LaTeX provides the following specialized math font commands.

Math font class	Package	Macro	Example	Result
Calligraphic	`amsmath`	`\mathcal`	`\mathcal{A,B,C}`	$\mathcal{A}, \mathcal{B}, \mathcal{C}$
Blackboard bold	`amssymb`	`\mathbb`	`\mathbb{R,N,Z}`	$\mathbb{R}, \mathbb{N}, \mathbb{Z}$
Double stroke	`dsfont`	`\mathds`	`\mathds{P,Q,R}`	$\mathbb{P}, \mathbb{Q}, \mathbb{R}$
Fraktur/Gothic	`eufrak`	`\mathfrak`	`\mathfrak{g,H,K}`	$\mathfrak{g}, \mathfrak{H}, \mathfrak{K}$

3 Alignment of mathematical expressions

The amsmath package provides specialized environments that allow us to structure sophisticated mathematical expressions by splitting them across multiple lines and aligning multiple equations in grids.

To use the amsmath package, we include the following line in the preamble:

```
\usepackage{amsmath}
```

3.1 Splitting an equation into multiple lines

Suppose we have a rather long equation whose length exceed the page width, such as that using the code below:

```
\begin{equation}
  (a+b)^{12}=a^{12}+12a^{11}b+66a^{10}b^2+220a^9b^3+495a^8b^4+792a^7b^5+924a^6b
^6+792a^5b^7+495a^4b^8+220a^3b^9+66a^2b^{10}+12ab^{11}+b^{12}
\end{equation}
```

which yields the following output that exceeds the margins of the page.

$$(a+b)^{12} = a^{12}+12a^{11}b+66a^{10}b^2+220a^9b^3+495a^8b^4+792a^7b^5+924a^6b^6+792a^5b^7+495a^4b^8+220a^3b^9+ \tag{1}$$

In such scenario, the multline environment can be used, where the equation can be split using double back-slash (\\). In this environment, the first line is left-aligned, the final line is right-aligned and intermediate lines are centered. The above long equation can be re-written as shown below:

```
\begin{multline}
  (a+b)^{12}=a^{12}+12a^{11}b \\
  +66a^{10}b^2+220a^9b^3+495a^8b^4+792a^7b^5 \\
  +924a^6b^6+792a^5b^7+495a^4b^8+220a^3b^9 \\
  +66a^2b^{10}+12ab^{11}+b^{12}
\end{multline}
```

which gives the following output

$$\begin{aligned}(a+b)^{12} = a^{12} + 12a^{11}b \\ &+ 66a^{10}b^2 + 220a^9b^3 + 495a^8b^4 + 792a^7b^5 \\ &+ 924a^6b^6 + 792a^5b^7 + 495a^4b^8 + 220a^3b^9 \\ &+ 66a^2b^{10} + 12ab^{11} + b^{12}\end{aligned} \tag{1}$$

However if we want all the lines of the equation to be centre aligned instead, then we can use the `gathered` environment wrapped within an outer `equation` environment, as shown below:

```
\begin{equation}
  \begin{gathered}
    (a+b)^{12}=a^{12}+12a^{11}b \\
    +66a^{10}b^2+220a^9b^3+495a^8b^4+792a^7b^5 \\
    +924a^6b^6+792a^5b^7+495a^4b^8+220a^3b^9 \\
    +66a^2b^{10}+12ab^{11}+b^{12}
  \end{gathered}
\end{equation}
```

which yields the following output:

$$\begin{gathered}(a+b)^{12}=a^{12}+12a^{11}b \\ +66a^{10}b^2+220a^9b^3+495a^8b^4+792a^7b^5 \\ +924a^6b^6+792a^5b^7+495a^4b^8+220a^3b^9 \\ +66a^2b^{10}+12ab^{11}+b^{12}\end{gathered} \tag{1}$$

3.2 Aligning multiple equations

The most effective way to collect and align multiple equations is by using the `align` environment. In this environment, the expressions in different lines are aligned at specified horizontal anchor points marked by ampersands (`&`). The lines are broken into rows using double back-slashes (`\\`).

For example, the long equation from the preceding sub-section can be split and aligned using the `align` environment as shown below:

```
\begin{align}
  (a+b)^{12}&=a^{12}+12a^{11}b \\
  &+66a^{10}b^2+220a^9b^3+495a^8b^4+792a^7b^5 \\
  &+924a^6b^6+792a^5b^7+495a^4b^8+220a^3b^9 \\
  &+66a^2b^{10}+12ab^{11}+b^{12}
\end{align}
```

which yields the following output:

$$\begin{aligned}(a+b)^{12}&=a^{12}+12a^{11}b && (1)\\ &+66a^{10}b^2+220a^9b^3+495a^8b^4+792a^7b^5 && (2)\\ &+924a^6b^6+792a^5b^7+495a^4b^8+220a^3b^9 && (3)\\ &+66a^2b^{10}+12ab^{11}+b^{12} && (4)\end{aligned}$$

As it can be observed, the `align` environment numbers each line with unique equation numbers. Such numbering can be suppressed for all the lines by using the `align*` instead, as shown below:

```
\begin{align*}
  (a+b)^{12}&=a^{12}+12a^{11}b \\
  &+66a^{10}b^2+220a^9b^3+495a^8b^4+792a^7b^5 \\
  &+924a^6b^6+792a^5b^7+495a^4b^8+220a^3b^9 \\
  &+66a^2b^{10}+12ab^{11}+b^{12}
\end{align*}
```

which gives the output:

$$\begin{aligned}(a+b)^{12} &= a^{12}+12a^{11}b \\ &+66a^{10}b^2+220a^9b^3+495a^8b^4+792a^7b^5 \\ &+924a^6b^6+792a^5b^7+495a^4b^8+220a^3b^9 \\ &+66a^2b^{10}+12ab^{11}+b^{12}\end{aligned}$$

In case we wish to suppress the numbering only on specific lines, then in the `align` environment we can use the `\nonumber` macro just before the `\\` for that line. For example, consider the following code:

```
\begin{align}
  (a+b)^{12}&=a^{12}+12a^{11}b \nonumber\\
  &+66a^{10}b^2+220a^9b^3+495a^8b^4+792a^7b^5 \\
  &+924a^6b^6+792a^5b^7+495a^4b^8+220a^3b^9 \nonumber\\
  &+66a^2b^{10}+12ab^{11}+b^{12} \nonumber
\end{align}
```

which gives the following output:

$$\begin{aligned}(a+b)^{12} &= a^{12}+12a^{11}b \\ &+66a^{10}b^2+220a^9b^3+495a^8b^4+792a^7b^5 \\ &+924a^6b^6+792a^5b^7+495a^4b^8+220a^3b^9 \\ &+66a^2b^{10}+12ab^{11}+b^{12}\end{aligned} \tag{1}$$

4 Typesetting matrices in LaTeX

The `amsmath` package provides commands to typeset matrices with different delimiters. Once the `amsmath` package has been included in the preamble, different types of matrices can be typeset using different environments. Each of these environments have to be wrapped inside an outer math environment, such as `equation` and `align`.

The general syntax is shown below:

```
  \begin{matrix environment name}
    {row 1, col 1} & {row 1, col 2} & {...} & {row 1, col n} \\
    {row 2, col 1} & {row 2, col 2} & {...} & {row 2, col n} \\
    {...} & {...} & {...} & {...} \\
    {row m, col 1} & {row m, col 2} & {...} & {row m, col n}
  \end{matrix environment name}
```

4.1 Plain matrix

A *plain matrix* is one in which the elements are arranged in rows and columns without any enclosing delimiters. For instance, the following code is used to typeset a plain matrix:

```
\begin{equation*}
  \begin{matrix}
    {1} & {-9} & {5} \\
    {4} & {8} & {-2} \\
    {-6} & {3} & {7}
  \end{matrix}
\end{equation*}
```

which gives the following output:

$$\begin{matrix} 1 & -9 & 5 \\ 4 & 8 & -2 \\ -6 & 3 & 7 \end{matrix}$$

4.2 Parentheses matrix

A *parentheses matrix* is one in which the elements are arranged in rows and columns, enclosed within variable sized parentheses `( )` as delimiters. For instance, the following code is used to typeset a parentheses matrix:

```
\begin{equation*}
  \begin{pmatrix}
    {1} & {-9} & {5} \\
    {4} & {8} & {-2} \\
    {-6} & {3} & {7}
  \end{pmatrix}
\end{equation*}
```

which gives the following output:

$$\begin{pmatrix} 1 & -9 & 5 \\ 4 & 8 & -2 \\ -6 & 3 & 7 \end{pmatrix}$$

4.3 Brackets matrix

A *brackets matrix* is one in which the elements are arranged in rows and columns, enclosed within variable sized brackets `[ ]` as delimiters. For instance, the following code is used to typeset a brackets matrix:

```
\begin{equation*}
  \begin{bmatrix}
    {1} & {-9} & {5} \\
    {4} & {8} & {-2} \\
    {-6} & {3} & {7}
  \end{bmatrix}
\end{equation*}
```

which gives the following output:

$$\begin{bmatrix} 1 & -9 & 5 \\ 4 & 8 & -2 \\ -6 & 3 & 7 \end{bmatrix}$$

4.4 Braces matrix

A *braces matrix* is one in which the elements are arranged in rows and columns, enclosed within variable sized braces `{ }` as delimiters. For instance, the following code is used to typeset a braces matrix:

```
  \begin{equation*}
    \begin{Bmatrix}
      {1} & {-9} & {5} \\
      {4} & {8} & {-2} \\
      {-6} & {3} & {7}
    \end{Bmatrix}
  \end{equation*}
```

which gives the following output:

$$\begin{Bmatrix} 1 & -9 & 5 \\ 4 & 8 & -2 \\ -6 & 3 & 7 \end{Bmatrix}$$

4.5 Pipes matrix

A *pipes matrix* is one in which the elements are arranged in rows and columns, enclosed within variable sized pipes | | as delimiters. For instance, the following code is used to typeset a pipes matrix:

```
\begin{equation*}
  \begin{vmatrix}
    {1} & {-9} & {5} \\
    {4} & {8} & {-2} \\
    {-6} & {3} & {7}
  \end{vmatrix}
\end{equation*}
```

which gives the following output:

$$\begin{vmatrix} 1 & -9 & 5 \\ 4 & 8 & -2 \\ -6 & 3 & 7 \end{vmatrix}$$

4.6 Double-pipes matrix

A *double pipes matrix* is one in which the elements are arranged in rows and columns, enclosed within variable sized double-pipes || || as delimiters. For instance, the following code is used to typeset a double-pipes matrix:

```
\begin{equation*}
  \begin{Vmatrix}
    {1} & {-9} & {5} \\
    {4} & {8} & {-2} \\
    {-6} & {3} & {7}
  \end{Vmatrix}
\end{equation*}
```

which gives the following output:

$$\begin{Vmatrix} 1 & -9 & 5 \\ 4 & 8 & -2 \\ -6 & 3 & 7 \end{Vmatrix}$$

5 LaTeX packages for Physics

5.1 The `siunitx` package

The `siunitx` package provides tools for typesetting physical quantities, numbers, and units consistently in LaTeX. Maintained by Joseph Wright, the current version (v3) provides extensive automated formatting, parsing, rounding, and decimal-aligned table columns.

To use the `siunitx` package, we include the following line in the preamble:

```
\usepackage{siunitx}
```

The `siunitx` package enforces the BIPM (International Bureau of Weights and Measures) standards for formatting numbers and units. For physics lab reports and dissertations, it prevents critical formatting errors such as italicizing units, misaligning decimals or using incorrect spacing between values and units.

5.1.1 Fundamental macros

The fundamental macros provided by `siunitx` handles the spacing as well as the font selection. These are briefly summarised below:

1. For standalone numbers

5.2 The `physics` package

The `physics` package is a popular LaTeX package created by Sergio C. de la Barrera that simplifies and standardizes the typesetting of physics, vector calculus and linear algebra equations.

To use the `physics` package, we include the following line in the preamble:

```
\usepackage{physics}
```